Customer Segmentation System

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report **"Customer Segmentation System Using K-Means Clustering"** is the bonafide work of **GOPI KRISHNAN L (230701096) HARSHA VARDHANAN V (230701108)** who carried out the project work under my supervision.

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ABSTRACT

Customer segmentation plays an important role in modern business analytics by helping organizations understand customer behavior and improve decision-making. This project presents a full-stack Customer Segmentation System that uses Machine Learning techniques to classify customers into different groups based on purchasing behavior and customer attributes. The system is designed using K-Means clustering to automatically identify meaningful customer segments and provide actionable business insights. The application consists of a FastAPI backend and an interactive frontend dashboard developed using HTML, CSS, and JavaScript. The system processes customer data containing attributes such as income, spending score, purchase frequency, recency, and monetary value.

The proposed system provides visualization of customer clusters through interactive scatter plots and displays detailed segmentation insights such as VIP Customers, Churn Risk Customers, Loyal Budget Shoppers, and Impulsive Buyers. The application also offers recommendations for each customer segment to support marketing and customer retention strategies. By automating customer analysis and segmentation, the system reduces manual effort and enables businesses to better understand customer patterns. The project demonstrates a practical implementation of machine learning for customer analytics and business intelligence.

1. INTRODUCTION

Customer segmentation is an important technique used in business analytics and marketing to divide customers into groups based on similar characteristics and purchasing behavior. Understanding customer behavior helps organizations improve customer satisfaction, optimize marketing strategies, and increase business profitability. Traditional customer analysis methods are often time-consuming and may not effectively identify hidden patterns in large datasets. Machine learning techniques provide an efficient solution for analyzing customer data and generating meaningful customer segments automatically.

This project proposes a Customer Segmentation System using K-Means Clustering to classify customers into multiple groups based on factors such as income, spending score, purchase frequency, recency, and monetary value. The system uses unsupervised machine learning techniques to identify behavioral similarities among customers without requiring predefined labels.

The application is developed as a full-stack web application consisting of a FastAPI backend and a frontend dashboard built using HTML, CSS, and JavaScript. The backend handles data processing, feature engineering, clustering, and API communication, while the frontend provides interactive visualizations and user-friendly interfaces for displaying segmentation results.

The main objective of this project is to simplify customer analysis and provide businesses with actionable insights through automated customer segmentation. The system also improves decision-making by visualizing customer clusters and generating business recommendations for each segment. Overall, the project demonstrates the practical application of machine learning in customer relationship management and business intelligence.

2. LITERATURE REVIEW

Han and Kamber (2011) in their work “Data Mining: Concepts and Techniques” explained various data mining methods used for customer analysis and pattern discovery. Their research highlighted clustering techniques as effective approaches for grouping similar customer data and extracting useful business insights.
MacQueen (1967) introduced the K-Means clustering algorithm, one of the most widely used unsupervised machine learning techniques for customer segmentation. The algorithm partitions data into clusters by minimizing the distance between data points and cluster centroids. This method became a foundation for many modern clustering applications.

Jain (2010) in the paper “Data Clustering: 50 Years Beyond K-Means” discussed the importance and effectiveness of clustering algorithms in data analysis. The study emphasized the simplicity, scalability, and practical applications of K-Means clustering in business intelligence and customer analytics. Wedel and Kamakura (2012) in their research on market segmentation explained how customer grouping techniques help businesses identify target audiences and improve marketing efficiency. Their study showed that segmentation improves customer satisfaction and increases revenue through personalized services.

Pedregosa et al. (2011) introduced Scikit-learn, a powerful machine learning library for Python that provides efficient implementations of clustering algorithms, including K-Means. The library simplified machine learning development and became widely used in academic and industrial applications.
Aggarwal (2015) in the book “Data Mining: The Textbook” explained advanced clustering and customer analytics techniques used in modern recommendation and marketing systems. The study highlighted the role of machine learning in improving customer understanding and predictive analytics.

Recent developments in web-based machine learning systems have enabled the integration of interactive dashboards and visualization tools for customer analytics. Modern frameworks such as FastAPI and Chart.js provide scalable solutions for developing real-time customer segmentation applications with efficient data visualization and API communication.

3. PROPOSED SYSTEM

The proposed system is a web-based Customer Segmentation Application designed to analyze customer behavior using machine learning techniques. The system automatically groups customers into different clusters based on their purchasing patterns and behavioral attributes. The main objective of the system is to help businesses understand customer categories and improve marketing strategies through data-driven insights.

The application uses the K-Means clustering algorithm to segment customers into six distinct groups. Customer data such as income, spending score, purchase frequency, recency, and monetary value are used as input features for clustering. Additional engineered features including value score, engagement score, efficiency score, and loyalty score are generated to improve clustering accuracy and customer analysis.

The system consists of two major components: a FastAPI backend and a frontend dashboard developed using HTML, CSS, and JavaScript. The backend handles dataset loading, preprocessing, feature engineering, clustering, and API operations. The frontend provides an interactive dashboard that displays customer segmentation results through charts and visualizations.

The workflow of the system begins with loading the customer dataset. The data is then preprocessed and transformed into suitable numerical features for clustering. The K-Means algorithm groups customers into clusters based on similarities in behavioral patterns. Each cluster is mapped to meaningful business-oriented segment names such as VIP Customer, Churn Risk Customer, Loyal Budget Shopper, and Impulsive Buyer.

3.1 ARCHITECTURE DIAGRAM

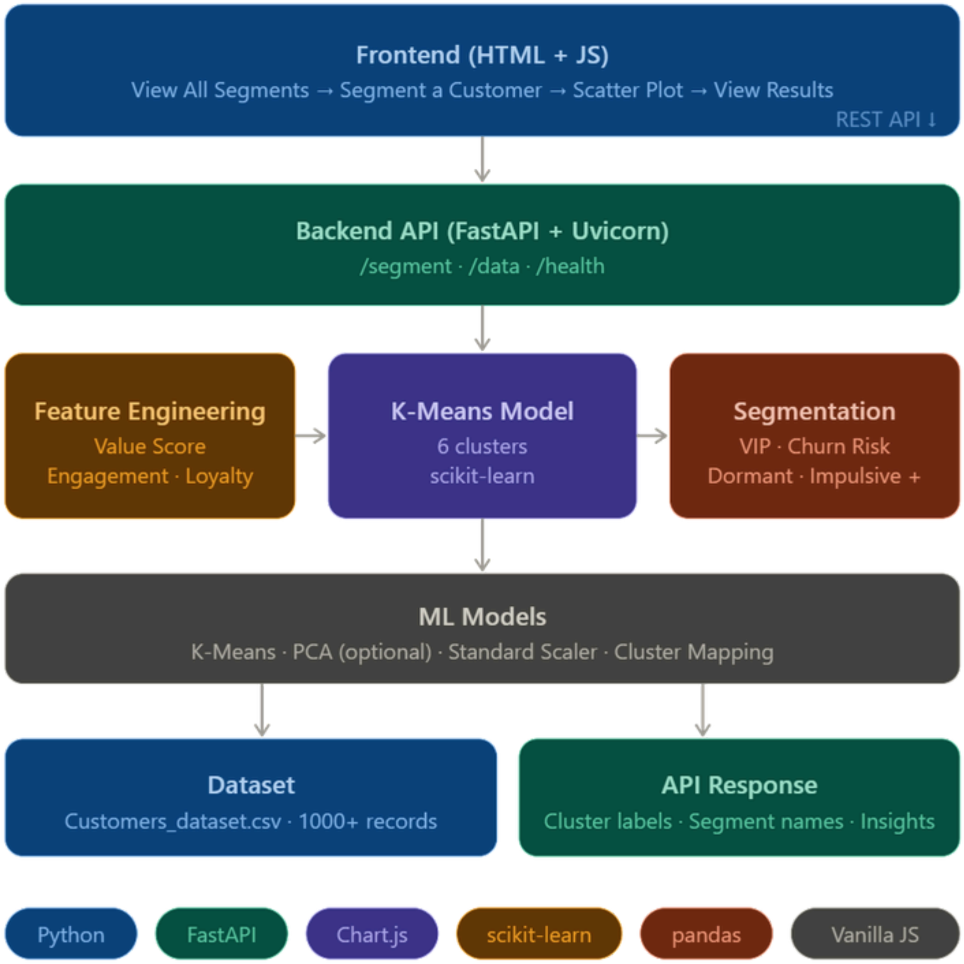


Figure 3.1 : Architecture

4. MODULES DESCRIPTION

4.1 MODULE 1 - Data Processing and Feature Engineering

This module is responsible for loading and preprocessing the customer dataset before clustering. The dataset is imported from a CSV file containing customer information such as age, gender, annual income, spending score, purchase frequency, recency, and monetary value.

The system first validates the dataset and checks for missing or inconsistent values. Data preprocessing techniques such as cleaning, normalization, and formatting are applied to ensure accurate analysis. Numerical values are standardized to improve clustering performance and maintain consistency between features.

Feature engineering is then performed to generate additional behavioral metrics from the existing customer data. Important engineered features include:

- Value Score = Income × Spending Score
- Engagement Score = Frequency × Monetary Value
- Efficiency Score = Spending Score / Age
- Loyalty Score = 1 / Recency

These engineered features help the system better understand customer behavior patterns and improve the quality of segmentation.

After preprocessing and feature engineering, the prepared dataset is passed to the clustering module for customer segmentation. This module ensures that the customer data is properly structured and optimized for machine learning analysis.

4.2 MODULE 2 - Customer Segmentation and Visualization

This module performs customer segmentation using the K-Means clustering algorithm. The system processes customer features such as income, spending score, frequency, recency, and monetary value to group customers into different clusters based on behavioral similarities.

After clustering, each customer group is assigned meaningful segment names such as VIP Customer, Churn Risk Customer, Loyal Budget Shopper, and Impulsive Buyer. Interactive visualizations including scatter plots and charts are generated to display customer distributions and cluster relationships.

The module also provides business insights and recommendations for each customer segment. The frontend dashboard communicates with the FastAPI backend through REST APIs to display segmentation results dynamically. This module automates customer analysis and helps businesses understand customer behavior effectively.

5. IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The Customer Segmentation System is implemented as a full-stack web application using machine learning and visualization technologies. The development environment and tools used for the implementation are as follows:

- Programming Language: Python
- Backend Framework: FastAPI
- Frontend Technologies: HTML, CSS, JavaScript
- Machine Learning Library: Scikit-learn
- Visualization Library: Chart.js
- Data Processing Libraries: Pandas, NumPy
- Server: Uvicorn

The system uses a customer dataset stored in CSV format containing more than 1000 customer records. The dataset includes attributes such as Customer ID, Gender, Age, Income, Spending Score, Frequency, Recency, and Monetary Value.

The implementation process includes the following stages:

1. Dataset loading and preprocessing
2. Feature engineering and normalization
3. K-Means clustering model training
4. Customer segment mapping
5. REST API integration
6. Interactive visualization generation

The backend API provides endpoints for customer segmentation, customer data retrieval, and visualization support. The frontend dashboard interacts with these APIs to display segmentation results and customer insights dynamically.

5.2 RESULTS

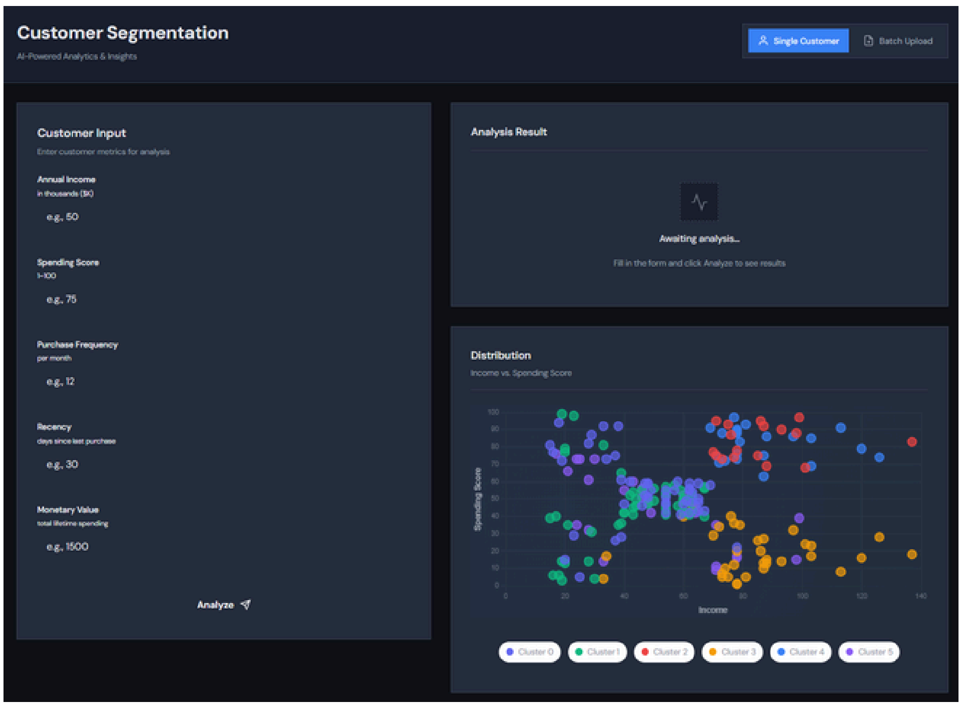


Figure 5.1 : Output 1

This output shows the homepage of the Customer Segmentation System. The interface provides options to visualize customer clusters and analyze customer behavior using machine learning techniques. Users can interact with the dashboard to access segmentation features and customer insights.

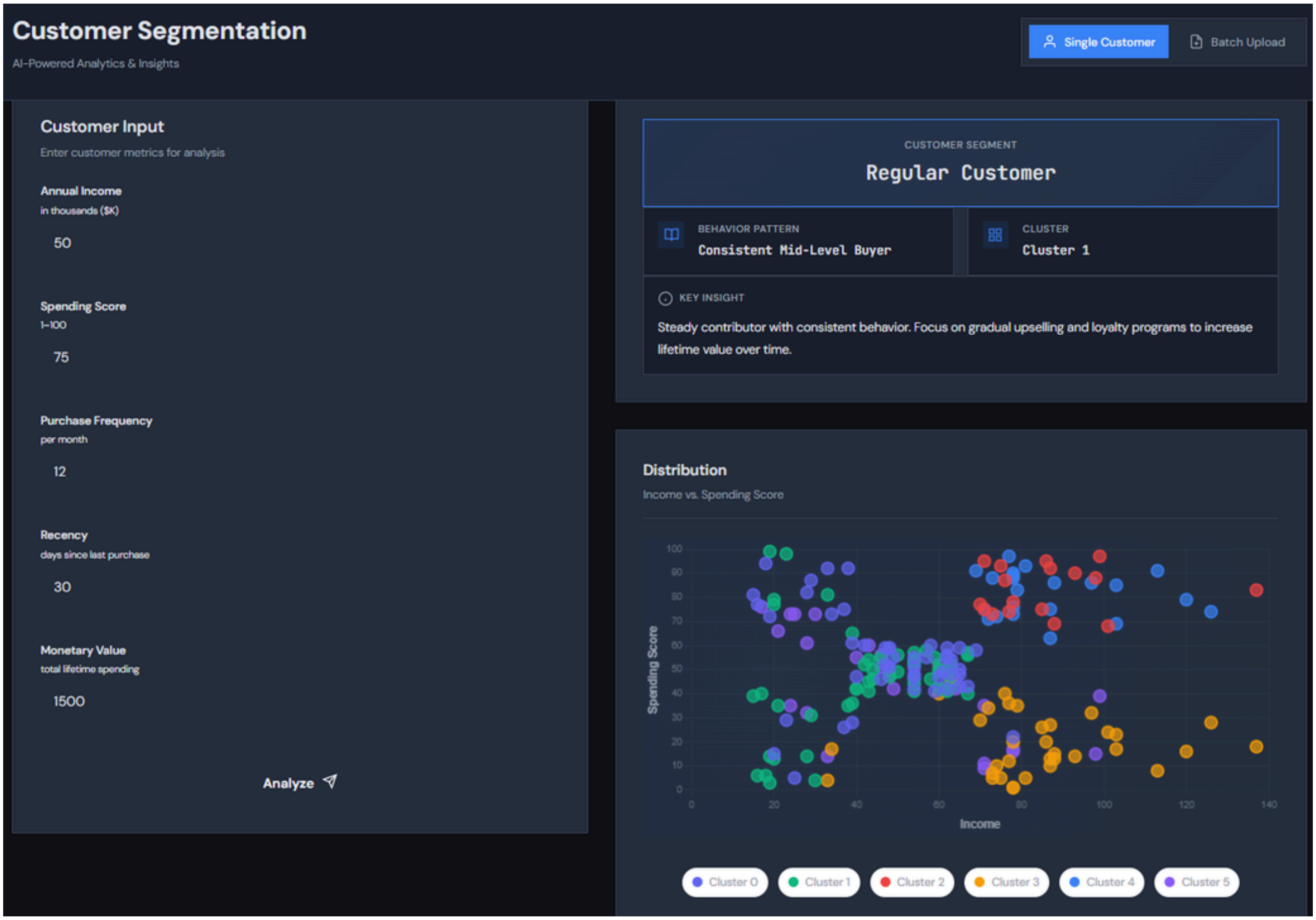
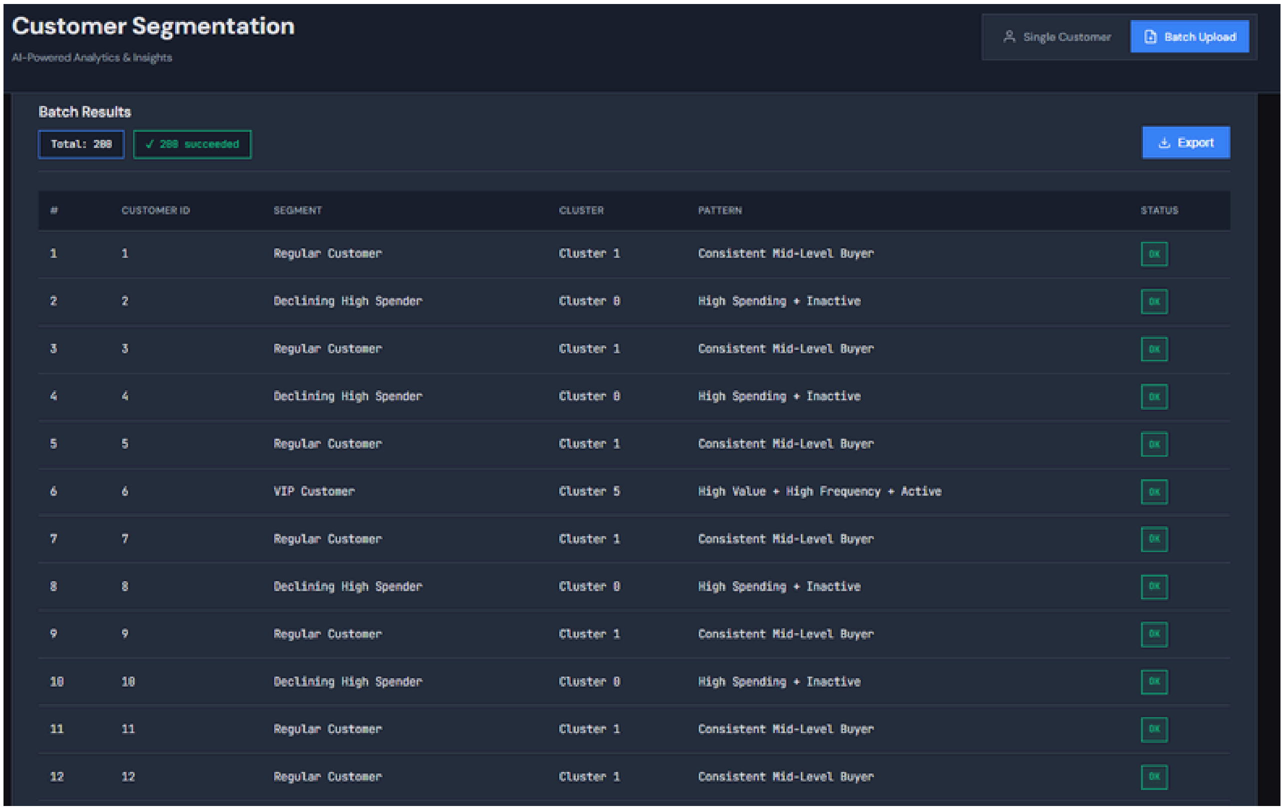


Figure 5.2 : Output 2

This output displays the customer segmentation visualization generated using the K-Means clustering algorithm. Customers are grouped into different clusters based on features such as income and spending score. Different colors are used to represent different customer segments in the scatter plot.



#	CUSTOMER ID	SEGMENT	CLUSTER	PATTERN	STATUS
1	1	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK
2	2	Declining High Spender	Cluster 0	High Spending + Inactive	OK
3	3	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK
4	4	Declining High Spender	Cluster 0	High Spending + Inactive	OK
5	5	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK
6	6	VIP Customer	Cluster 5	High Value + High Frequency + Active	OK
7	7	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK
8	8	Declining High Spender	Cluster 0	High Spending + Inactive	OK
9	9	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK
10	10	Declining High Spender	Cluster 0	High Spending + Inactive	OK
11	11	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK
12	12	Regular Customer	Cluster 1	Consistent Mid-Level Buyer	OK

Figure 5.3 : Output 3

This output shows the detailed customer segmentation results and business insights generated by the system. Each cluster is assigned a meaningful segment name such as VIP Customer, Churn Risk Customer, or Loyal Budget Shopper along with recommendations for customer engagement and retention strategies.

6. CONCLUSION AND FUTURE WORK

The Customer Segmentation System successfully applies machine learning techniques to analyze customer behavior and group customers into meaningful segments. The system automates customer analysis using the K-Means clustering algorithm and provides interactive visualizations along with business insights. By identifying customer groups such as VIP Customers, Churn Risk Customers, and Loyal Budget Shoppers, the system helps businesses improve marketing strategies, customer retention, and decision-making processes.

The integration of FastAPI, Scikit-learn, and interactive frontend technologies enables efficient data processing and real-time visualization of customer clusters. The project demonstrates the practical application of unsupervised machine learning in business analytics and customer relationship management.

In the future, the system can be enhanced by integrating advanced clustering algorithms and deep learning techniques for improved segmentation accuracy. Additional features such as real-time customer analytics, predictive recommendation systems, cloud deployment, and user authentication can also be implemented. Support for larger datasets and advanced visualization dashboards may further improve the scalability and usability of the system.

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